

Ngoc-Dung Nguyen

AI / MACHINE LEARNING RESEARCHER

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AI/ML researcher with a rare combination of **deep-learning engineering** and a **biomedical & cancer-science foundation**. Four years building predictive models on large clinical and prospective cohort data, including **peer-reviewed neural-network survival analysis for cancer risk** (Cancers, 2023). Hands-on with **PyTorch, Transformers, and deep survival models**, plus a strong statistical and medical-domain background. Seeking to apply biomedical AI to **AI-driven drug discovery**—translating molecular and clinical data into predictive, decision-ready models.

PROFESSIONAL EXPERIENCE

AI Researcher — *MoAdata* Jan 2022 – Present

- Develop **deep-learning models** that predict cancer risk from large-scale prospective cohort data, from data pipeline through training and evaluation.
- Built and open-sourced **nDeep**, a deep survival-analysis model for time-to-event risk prediction (github.com/ngocdung03/nDeep).
- Collaborate with AI engineers on **model deployment** and productionization of predictive models.
- Continuously upskill in modern deep learning (Transformers, CNNs, LSTMs) through the AI VIETNAM AIO2025 program.

Public Health / Data Researcher — *National Cancer Center, South Korea* May 2020 – Dec 2021

- Applied **survival analysis and machine-learning** techniques to prospective epidemiological data to model cancer risk, resulting in peer-reviewed publications.
- Integrated and cleaned large biomedical datasets; built statistical-analysis plans, Cox proportional-hazards models, and reproducible R/Python pipelines.
- Produced conference posters and manuscripts; presented at an international conference (APHA, 2018).

Researcher — *Vietnam Institute of Applied Medicine* Jun 2016 – Aug 2017

- Designed research plans and clinical-trial questionnaires; contributed to two research projects and two publications.

Research Assistant — *National COPD Prevention Project, Bach Mai Hospital* Jan 2016 – Jun 2016

- Built study questionnaires and collected patient data; led data-quality management—supervised interviews and cleaned raw data.

SELECTED PUBLICATIONS

- 2023** Chulyoung B., Boseon K., Sunha J., Jonghoon L., **Dung NN**. “A Study on Survival Analysis Methods Using **Neural Networks** to Prevent Cancers.” *Cancers*. Role: deep survival modeling, data cleaning, analysis, manuscript.
- 2021** **Dung NN**, Jinhee K., Mikyung K. “Metabolic health and central obesity are significantly associated with an increased risk of thyroid cancer: Data from the Korean Genome and Epidemiology Study.” *Cancer Epidemiology, Biomarkers & Prevention*. Role: research questions, data cleaning, survival analysis, manuscript.
- 2018** Huy NV, Mai DL, Thanh NV, **Dung NN**, Anh NT. “A Systematic Review of Effort–Reward Imbalance among Health Workers.” *Int. Journal of Health Planning and Management*.
- 2017** Huy NV, **Dung NN**, Hanh LT, Thanh NV. “Patient satisfaction with health-care services at a national institute of ophthalmology.” *Int. Journal of Health Planning and Management*. Role: data collection, logistic-regression analysis, manuscript.
- 2016** Huong LT, ..., **Dung NN**, et al. “Cigarette Smoking among Adolescents aged 13–15 in Viet Nam: Results from GYTS 2014 Data.” *Asian Pacific Journal of Cancer Prevention*. Role: logistic-regression analysis, manuscript.

TECHNICAL SKILLS

- Deep Learning**
PyTorch, Transformers, LSTMs, CNNs, deep survival analysis
- ML & Statistics**
scikit-learn, survival analysis (Cox PH), logistic regression, predictive modeling, TensorFlow
- Programming**
Python, R, SQL, STATA, ArcGIS
- Tools & Platforms**
Git, AWS, Docker, Kubernetes, Flask, Jupyter
- Domain**
Cancer epidemiology, cohort/clinical data, medical prognosis, biostatistics

EDUCATION

- MSc, Financial Engineering**
WorldQuant University, USA
2021–2024 · GPA 91/100
- MSc, Public Health**
NCC Graduate School of Cancer Science & Policy, South Korea
2017–2019 · GPA 4.12/4.30
- BSc, Public Health**
Hanoi Medical University, Vietnam
2011–2015 · GPA 7.64/10

LANGUAGES

- English IELTS 7.0
- Korean KIIP Level 5
- Vietnamese Native

RELEVANT CERTIFICATIONS

- AI for Medical Prognosis — *deeplearning.ai*
- AWS Machine Learning Fundamentals — *Udacity*
- Deep Learning with TensorFlow — *Codecademy*
- Docker & Kubernetes
- Advanced Statistical Methods in Python
- Mathematics for Machine Learning — *Imperial College London*
- Predictive Analytics Nanodegree — *Udacity*

Conference: APHA Annual Meeting & Expo, 2018 — HPV cohort & progression of cervical intraepithelial neoplasia (survival analysis).

SELECTED PROJECTS

nDeep — Deep Survival Analysis Model

- ▶ Deep-learning model for time-to-event cancer-risk prediction on prospective cohort data; neural network for survival analysis. github.com/ngocdung03/nDeep

Machine-Learning Risk-Factor Analysis (NCC)

- ▶ Applied ML algorithms in R/Python to identify predictive factors of disease from metabolic and epidemiological data, contributing to a peer-reviewed study.

Deep-Learning Specialization & Applied ML Portfolio

- ▶ Hands-on implementations across CNNs, RNNs/LSTMs, and TensorFlow/PyTorch model building. github.com/ngocdung03

PROFESSIONAL DEVELOPMENT

- ▶ **AI VIETNAM — AIO2025**, intensive AI/ML program (2025)
- ▶ Kaggle Masterclass; SQL & Python data-science tracks
- ▶ Good Clinical Practice; Writing in the Sciences (Stanford)

AWARDS

2018 · Overseas Academic Conference Scholarship, National Cancer Center (Korea)

2015 · Active Student of the Whole Course, Hanoi Medical University

2011–12 · Total Scholarship, Hanoi Medical University